

Technical Document #84: Software Engineering - Comprehensive Overview

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Continuous integration (CI) automatically builds and tests code changes. It catches integration issues early in the development process.

Continuous deployment (CD) automatically deploys code changes to production. It enables rapid, reliable releases.

Technical debt represents the cost of choosing easy solutions over better ones. It accumulates over time and slows development velocity.

API design defines how software components communicate. RESTful APIs and GraphQL are popular API design approaches.

Refactoring improves code structure without changing behavior. It makes code easier to understand, maintain, and extend.

Test-driven development (TDD) writes tests before writing code. It improves code quality and design through small, incremental changes.

Autonomous vehicles use a combination of sensors, AI, and mapping to navigate without human intervention. Self-driving technology is rapidly advancing.

Augmented reality overlays digital information on the physical world. It has applications in training, maintenance, and entertainment.

Supply chain management leverages technology to optimize logistics and distribution. Real-time tracking and predictive analytics improve efficiency.

Edge computing processes data closer to its source, reducing latency and bandwidth usage. It is essential for real-time applications like autonomous vehicles.

Key Takeaways:

- Continuous deployment (CD) automatically deploys code changes to production.
- Code review examines code changes before merging.
- Microservices architecture structures applications as independently deployable services.